



FLATLAND

EDWIN A. ABBOTT

Preface to the Second and Revised Edition, 1884

By the Editor If my poor Flatland friend retained the vigour of mind which he enjoys when he began to compose these Memoirs, I should not now need to represent him in this preface, in which he desires, firstly, to return his thanks to his readers and critics in Spaceland, whose appreciation has, with unexpected celerity, required a second edition of his work; secondly, to apologize for certain errors and misprints (for which, however, he is not entirely responsible); and, thirdly, to explain one or two misconceptions. But he is not the Square he once was. Years of imprisonment in the still heavier burden of general incredulity and mockery, have combined with the natural decay of old age to erase from his mind many of the thoughts and notions, and much also of the terminology, which he acquired during his short stay in Spaceland. He has, therefore, requested me to reply in his behalf to two special objections, one of an intellectual, the other of a moral nature. The first objection is, that a Flatlander, seeing a Line, sees something that sh^{ou}ld be *thick* to the eye as well as long to the eye (otherwise it would not be visible, if it had not some thickness); and consequently he ought (it is argued) to acknowledge that his countrymen are not only long and broad, but also (though doubtless in a very slight degree) *thick* or *high*. This objection is plausible, and, to Spacelanders, almost irresistible, so that, I confess, when I first heard it, I knew not what to reply. But my poor old friend's answer appears to me completely to meet it. "I admit," said he,—when I mentioned to him this objection—"I admit the truth of your critic's facts, but I deny his conclusions. It is true that we have really in Flatland a Third unrecognized Dimension called 'height'; just as it is also true that you have really in Spaceland a Fourth unrecognized Dimension, called by no name at present, but which I will call 'extra-height.' But we can no more take cognizance of our 'extra-height' than you can of your 'extra-height.' Even I—who have been in Spaceland, and have had the privilege of understanding

Flatland

Edwin A. Abbott · 2018

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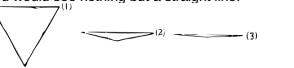
for twenty-four hours the meaning of 'height'—even I cannot now comprehend it, nor realize it by the sense of sight or by any process of reason; I can but apprehend it by faith. "The reason is obvious. Dimension implies direction, implies measurement, implies the more and the less. Now, all our lines are *equally* and *infinitesimally* thick (or high, whichever you like); consequently, there is nothing in them to lead our minds to the conception of that Dimension. No 'delicate micrometer'—as has been suggested by one too hasty Spaceland critic—would in the least avail us; for we should not know *what* to *measure*, *nor* in *what direction*. When we see a Line, we see something that is long and *bright*, *brightness*, as well as length, is necessary to the existence of a Line; if it had no length, it vanishes, the Line is extinguished. Hence, all my Flatland friends—when I talk to them about the unrecognized Dimension which is somehow visible in a Line—say, 'Ah, you mean *brightness*.' And when I reply, 'No, I mean a real Dimension,' they at once retort, 'Then measure it, or tell us in what direction it extends; and this silences me, for I can do neither. Only yesterday, when the Chief Circle (in other words the High Priest) came to inspect the state prison and paid me his seventh annual visit, and when for the seventh time he put me the question, 'Was I any better?' I tried to prove to him that he was 'high,' as well as long and broad, although he did not know it. But what was his reply? 'You say I am "high"; measure my "high-ness" and I will believe you.' What could I do? How could I meet his challenge? I was crushed; and he left the room triumphant. "Does this still seem strange to you? Then put yourself in a similar position. Suppose a person of the Fourth Dimension, condescending to visit you, were to say, 'Whenever you open your eyes, you see a Plane (which is of Two Dimensions) and you *infer* a Solid (which is of Three); but in reality you also see (though you do not recognize) a Fourth Dimension, which is not colour nor brightness nor anything of the kind, but a true Dimension, although I cannot point out to you its direction, nor can you possibly measure it.' What would

1a / 3 - 6% additional assistance; and in the towns we have the guidance of the houses, which of course have their sidewalks running for the most part North and South, so that the roofs may keep off the rain from the North. In the country, where there are no houses, the trunks of the trees serve as some sort of guide. Altogether, we have not so much difficulty as might be expected in determining our bearings. Yet in our more temperate regions, in which the southward attraction is hardly felt, walking sometimes in a perfectly desolate plain where there have been no houses nor trees to guide me, I have been occasionally compelled to remain stationary for hours together, waiting till the rain came before continuing my journey. On the weak and aged, and especially on delicate females, the force of attraction tells much more heavily than on the robust of the male sex, so that it is a point of breeding, if you meet a lady in the street, always to give her the North side of the way—by no means an easy thing to do always, since the weather is so variable, and the climate where it is difficult to tell your North from your South. Windows there are none in our houses: for the light comes to us alike in our homes and out of them, by day and by night, equally at all times and in all places, whence we know not. It was in old days, with our learned men, an interesting and oft-investigated question, "What is the origin of light?" and the solution of it has been repeatedly attempted, with no other result than to crowd the lunatic asylums with the would-be solvers. Hence, after fruitless attempts to suppress such investigations indirectly by making them liable to a heavy tax, the Legislature, in comparatively recent times, absolutely prohibited them. I—alas, I alone in Flatland—know now only too well the true solution of this mysterious problem; but my knowledge cannot be made intelligible to a single one of my countrymen; and I am mocked at—I, the sole possessor of the truths of Space and of the secrets of the world of Light from the world of the three Dimensions—as if I were the maddest of the mad! But a truce to these painful digressions: let me return to our houses. The most common form for the construction of a house is five-sided

"One touch of Nature makes all worlds akin." "I

On this point the defence of the Square seems to me to be impregnable. I wish I could say that his answer to the second (or moral) objection was equally clear and cogent. It has been objected that he is a woman-hater; and as this objection has been vehemently urged by those who the Nature's decree has constituted the somewhat larger half of the Spaceland race, it should like to remove it, so far as I can honestly do so. But the Square is so unaccustomed to the use of the moral terminology of Spaceland that I should be doing him an injustice if I were literally to transcribe his defence against this charge. Acting, therefore, as his interpreter and summarizer, I gather that in the course of an imprisonment of seven years he has himself modified his own personal views, both as regards women and as regards the *isosceles* or low classes. In a still more recent, it inclines to the opinion of the Sphere that the Straight Lines are in many important respects superior to the Circles. But, writing as a Historian, he has identified himself (perhaps too closely) with the views generally adopted by Flatland, and (as he has been informed) even by Spaceland, Historians; in whose pages (until very recent times) the destinies of women and of the masses of mankind have seldom been deemed worthy of mention and need of an new era of civilization. In a still more obscure passage he now desires to disavow the Circular or aristocratic tendencies with which some critics have naturally credited him. While doing justice to the intellectual power with which a few Circles have for many generations maintained

have said "my universe"; but now my mind has been opened to higher views of things. In such a country, you will perceive at once that it is impossible that there should be anything of what you call a "solid" kind; but I dare say you will suppose that we are not so stupid as to be so ignorant of our own nature. In other Figures, moving about as I have described them. On the contrary, we could see nothing of the kind, not at least so as to distinguish one Figure from another. Nothing was visible, nor could be visible, to us, except Straight Lines; and the necessity of this I will speedily demonstrate. Place a penny on the middle of one of your tables in Space; and leaning over it, look down upon it. It will appear a circle. But now, drawing back to the edge of the table, gradually lowering your eye, you will bring yourself more and more into the condition of the inhabitants of Flatland, and you will find the penny becoming more and more oval to your view, and at last when you have placed your eye exactly on the edge of the table (so that you are, as it were, actually a Flatlander) the penny will then have ceased to appear oval at all, and will have become, so far as you can see, a straight line. The same thing would happen if it were possible to take the table, gradually lowering your eye, and bring it out of the table. As soon as you look at it with your eye on the edge of the table, you will find that it ceases to appear to you a Figure, and that it becomes in appearance a straight line. Take for example an equilateral Triangle—who represents with us a tradesman of the respectable class—Fig. 1. represents the tradesman as you would see him while you were bending over him from above; figs. 2 and 3 represent the same Triangle, as you would see it when you were on the level, or all but on the level of the table; and if your eye were quite on the level of the table (and that is how we see him in Flatland) you would see nothing but a straight line.



1a / 3 - 10% again, for fear lest the freshly developed organism may, by force of unconscious imitation, fall back again into his hereditary level. The occasional emergence of an Equilateral from the ranks of his self-born ancestors is welcomed, not only by the poor serfs themselves, as a gleam of light and hope shed upon the monotonous squalor of their existence, but also by the aristocracy at large; for all the higher classes are well aware that these rare phenomena, while they do little or nothing to improve the condition of the lower classes, do constitute a useful barrier against revolution from below. Had the accursed rabble been all, without exception, absolutely destitute of hope and of ambition, they might have found leaders in some of their many seditious outbreaks, so able as to render their superior numbers and strength too much even for the wisdom of the Circles. But a wise ordinance of Nature has decreed that, in proportion as the working-classes increase in intelligence, knowledge, and all virtue, in that same proportion their acute angle (which makes them physically terrible) shall increase also and approximate to the comparatively harmless angle of the Equilateral Triangle. Thus, in the most brutal and formidable of the soldier class—creatures almost on a level with women in their lack of intelligence—it is found that, as they wax in the mental ability necessary to employ their tremendous penetrating power to advantage, so do they wax in the power of self-control, self-restraint, self-control, self-control! And how perfect a proof of the natural fitness and, I may almost say, the divine origin of the aristocratic constitution of the States in Flatland! By a judicious use of this law of Nature, the Polygons and Circles are almost always able to stifle sedition in its very cradle, taking advantage of the irrepressible and boundless hopefulness of the human mind. Art also comes to the aid of law and order. It is generally found that a little artificial compression of the body, as one of the part of the State physicians—to make some of the more intelligent leaders of a rebellion perfectly Regular, and to admit them at once into the privileged classes; a much larger

their supremacy over immense multitudes of their countrymen, he believes that the facts of Flatland, speaking for themselves without comment on his part, declare that Revolutions cannot always be suppressed by slaughter, and that Nature, in punishing the Circles, as well as punishing the Squares, is pointing to the ultimate failure—and herein," he says, "I see a fulfilment of the great law of all worlds, that while the wisdom of Man thinks it is working one thing, the wisdom of Nature constrains it to work another, and quite a different and far better thing." For the rest, he begs his readers not to suppose that every minute detail in the daily life of Flatland must needs correspond to some other detail in Spaceland; and yet he hopes that, taken as a whole, his work may prove suggestive, well as amusing, to those Spacelanders of moderate and modest minds who—speaking of that which is of the highest importance, but lies beyond experience—decline to say on the one hand, "This can never be," and on the other hand, "It must needs be precisely thus, and we know all about it."

1. The author desires me to add, that the misconception of some of his critics on this matter has induced him to insert in his diagram the Chief Circle, and to make certain remarks bearing on the point in question, and which he had previously omitted as being tedious and unnecessary.

[Sheet 1 · Front · Cell 3]

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Part I This World "Be patient, for the world is broad and wide."

I Of the Nature of Flatland

I call our world Flatland, not because we call it so, but to make its nature clear to my happy readers, who are privileged to live in Space. Imagine a vast sheet of paper on which straight Lines, Triangles, Squares, Pentagons, Hexagons, and other Figures, instead of remaining fixed in their places, move freely about, on or in the surface, but without the power of rising above or sinking below it, very much like shadows—only hard and with luminous edges—and you will then have a pretty correct notion of my country and countrymen. Alas, a few years ago, I should

When I was in Spaceland I heard that your sailors have very similar experiences while they traverse your seas and discern some distant island or coast lying on the horizon. The far-off land may have bays, peninsulas, angles in and out to any number of degrees, and the sun, moon, and stars may be seen to rise and indeed your sun shines bright upon them revealing the projections and retirements by means of light and shade), nothing but a grey unbroken line upon the water. Well, that is just what we see when one of our triangular or other acquaintances comes toward us in Flatland. As there is neither sun with us, nor any light of such a kind as to make shadows, we have none of the helps to the sight that you have in Spaceland. The friends of the Flatlander, however, as they become larger, if he leaves us it becomes smaller; but still he looks like a straight line; be he a Triangle, Square, Pentagon, Hexagon, Circle, what you will—a straight Line he looks and nothing else. You may perhaps ask how under these disadvantageous circumstances we are able to distinguish our friends from one another: but the answer to this very natural question will be more fully and easily given when I come to the description of the Flatlanders. For the present let me defer this subject, and say a word or two about the climate and houses in our country.

II Of the Climate and Houses in Flatland

As with you, so also with us, there are four points of the compass: North, South, East, and West. There being no sun nor other heavenly bodies, it is impossible for us to determine the north in the usual way; but we have a method of our own. By a law of attraction, the force of the Southward attraction to the South, and, although in temperate climates this is very slight—so that even a woman in reasonable health can journey several furlongs northward without much difficulty—yet the hampering effect of the southward attraction is quite sufficient to serve as a compass in most parts of our earth. Moreover, the rain (which falls at stated intervals) coming always from the North, is an

number, who are still below the standard, allured by the prospect of being ultimately ennobled, are induced to enter the State Hospitals, where they are kept in honourable confinement for life; one or two alone of the more obstinate, foolish, and hopelessly irregular are led to execution. Then the wretched rabble of the *isosceles*, planless and leaderless, are either transfixed without resistance by the small body of their brethren whom the Chief Circle keeps in pay for emergencies of this kind; or, if they are more numerous, they are jealously and suspiciously fomented among them by the Circular party, they are stirred to mutual warfare, and perish by one another's angles. No less than one hundred and twenty rebellions are recorded in our annals, besides minor outbreaks numbered at two hundred and thirty-five; and they have all ended thus.

2. "What need of a certificate?" a Spaceland critic may ask: "Is not the procreation of a Square upon a certificate from Nature herself, proving the entire correctness and wisdom of the law, that no lady of any position will marry an uncertified Triangle. Square offspring has sometimes resulted from a slightly irregular Triangle; but in almost every such case the irregularity of the first generation is visited on the third; which either fails to attain the Pentagonal rank, or relates to the Triangular." [Sheet 1 · Front · Cell 14]

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IV Concerning the Women

If our highly pointed Triangles of the soldier class are formidable, it may be readily inferred that far more formidable are our women. For if a soldier is a wedge, a woman is a needle; being, so to speak, all point, at least at the two extremities. Add to this the power of making herself practically invisible at will, and you will perceive that a female, in Flatland, is a creature by no means to be trifled with. But here, perhaps, some of my younger readers may ask, "How can a woman be invisible?" To this I can only answer, "This is a question which I think, to be apparent without any explanation. However, a few words will make it clear to the most

or any disease necessitating involuntary motions, shall be instantly destroyed.

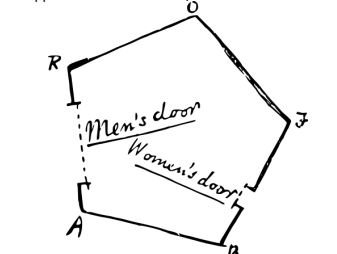
In some of the States there is an additional law forbidding females to order the execution of death, from walking or standing in any public place without moving their backs constantly from right to left so as to indicate their presence to those behind them; others oblige a woman, when travelling, to be followed by one of her sons, or servants, or by her husband; others confine women altogether to their houses except during the religious festivals. But it has been found by the wisest of our Circles or statesmen that the multiplication of restrictions on females tends not only to the annihilation of the race, but also to the increase of domestic murders to such an extent that a State loses more than it gains by a too prohibitive code. For whenever the temper of the women is thus exasperated by confinement at home or hampering regulations abroad, they are apt to vent their spleen upon their husbands and children; and in the less temperate climates the whole male population of a village has been sometimes destroyed in one or two hours of simultaneous male outbreak. Hence the Three Laws, mentioned above, suffice for the better regulated States, and may be accepted as a rough exemplification of our Female Code. After all, our principal safeguard is found, not in legislature, but in the interests of the women themselves. For, although they can inflict instantaneous death by a retrograde movement, yet unless they can at once disengage their stinging extremity from the struggling body of their victim, their own frail bodies are liable to be crushed by the annihilation of the race, but also to be out that in some less civilized States no female is suffered to stand in any public place without swaying her back from right to left. This practice has been universal among ladies of any pretensions to breeding in all well-governed States, as far back as the memory of Figures can reach. It is considered a disgrace to any State that legislation should have to enforce what ought to be, and is in every respectable female, a natural instinct. The back in our ladies of Circular rank is envied and imitated by the

At this period, square houses were still everywhere permitted, though discouraged by a special tax. But, about three centuries afterwards, the law decreed that in all towns containing a population above ten thousand, the angle of a Pentagon was the smallest house-angle that could be allowed consistently with the public safety. The good sense of the community has seconded the efforts of the Legislature; and now, even in the country, the pentagonal construction has superseded every other. It is only now and then in some very remote and backward agricultural district that an antiquarian may still discover a square house.

III Concerning the Inhabitants of Flatland

The greatest length or breadth of a full grown inhabitant of Flatland may be estimated at about eleven of your inches. Twelve inches may be regarded as a maximum. Our women are Straight Lines. Our soldiers and lowest classes of workmen are Triangles with two equal sides, each about eleven inches long, and a base or third side so short (often not exceeding half an inch) that they form at their vertices a very sharp and formidable angle. Indeed when their bases are of the most degraded type (not more than the eighth part of an inch in size), they can hardly be distinguished from Straight Lines or women; so extremely pointed are their vertices. With us, as with you, these Triangles are distinguished from others by being called "isosceles"; and by this name I shall refer to them in the following pages. Our middle class consists of Equilateral or Equal-Sided Triangles. Our professional men, and gentlemen are Squares (to which class I myself belong) and Five-Sided Figures or Pentagons. Next above these come the nobility, of whom there are several degrees, beginning at Six-Sided Figures, or Hexagons, and from thence rising in the number of their sides till they receive the honourable title of Polygonal, or many-sided. Finally when the number of the sides becomes so numerous, and the sides themselves so small, that the Figure cannot be distinguished from a Circle, he is included in the Circular or Priestly order; and this is the highest class of all. It is

or pentagonal, as in the annexed Figure. The two Northern sides RO, OF, constitute the roof, and for the most part have no doors; on the East is a small door for the women; on the West a much larger one for the men; the South side or floor is usually doorless. Square and triangular houses are not allowed, and for this reason. The angles of a Square (and still more those of an equilateral Triangle), being much more pointed than those of a Pentagon, and the lines of inanimate objects (such as houses) being dimmer than the lines of men and women, it follows that there is no little danger lest the points of a square or triangular house residence might do serious injury to an inconsiderate or perhaps absentminded traveller suddenly therefore, running against them; and as early as the eleventh century of our era, triangular houses were universally forbidden by law, the only exceptions being fortifications, powder-magazines, barracks, and other state buildings, which were exempted from the law, the general public should approach without circumspection.



a law of Nature with us that a male child shall have one more side than his father, so that each generation shall rise (as a rule) one step in the scale of development and nobility. Thus the son of a Square is a Pentagon; the son of a Pentagon, a Hexagon; and so on. But this rule applies not always to the tradesmen, and still less often to the soldiers, and to the workmen; who indeed can hardly be said to deserve the name of human Figures, since they have not all their sides equal. With them therefore the law of Nature does not hold; and the son of an *isosceles* (i.e. a Triangle with two sides equal) remains *isosceles* still. Nevertheless, as hope is not shut out, even from the *isosceles*, that his posterity may ultimately attain above his degraded condition. For, after a long series of military successes, or diligent and skilful labours, it is generally found that the more intelligent among the artisan and soldier classes manifest a slight increase of their third side or base, and a shrinkage of the two other sides. Inter marriages (arranged by the Priests) between the sons and daughters of these more intellectual members of the lower classes generally result in an offspring approximating still more to the type of the Equal-Sided Triangle. Rarely in proportion to the vast numbers of *isosceles* births—is a genuine and certifiable Equal-Sided Triangle produced from *isosceles* parents. 2 Such a birth requires, as its antecedents, not only a series of carefully arranged intermarriages, but also a long, continued exercise of frugality and self-control on the part of the would-be ancestors of the coming Equilateral, and a patient, systematic, and continuous education of the *isosceles* intellect through many generations. The birth of a True Equilateral Triangle from *isosceles* parents is the subject of rejoicing in our country for many furlongs around. After a strict examination conducted by the Sanitary and Social Board, the infant, if certified as Regular, is with solemn ceremonial admitted into the class of Equilaterals. He is then immediately taken from his proud yet sorrowful parents and adopted by some childless Equilateral, who is bound by oath never to permit the child henceforth to enter his former home or so much as to look upon his relations

unreflecting. Place a needle on a table. Then, with your eye on the level of the table, look at it sideways, and you see the whole length of it; but look at it endways, and you see nothing but a point. And so it is with us. We are all, in the eyes of one of our women. When her side is turned towards us, we see her as a straight line; when the end containing her eye or mouth—for with us these two organs are identical—is the part that meets our eye, then we see nothing but a highly lustrous point; but when the back is presented to our view, then—being only sub-lustrous, and, indeed, almost as dim as an inanimate object—her hinder extremity serves her as a kind of invisible cape. The danger to which we are exposed from this is only as a dim sub-lustrous point, how difficult must it be, even for the most cautious, always to avoid collision! Many are the enactments made at different times in the different States of Flatland, in order to minimize this peril; and in the Southern and less temperate climates where the force of gravitation is greater, and human beings more liable to casual and involuntary motions, the laws concerning women are naturally more strict. The danger to which we are exposed from this is only as a dim sub-lustrous point, how difficult must it be, even for the most cautious, always to avoid collision! Many are the enactments made at different times in the different States of Flatland, in order to minimize this peril; and in the Southern and less temperate climates where the force of gravitation is greater, and human beings more liable to casual and involuntary motions, the laws concerning women are naturally more strict. The danger to which we are exposed from this is only as a dim sub-lustrous point, how difficult must it be, even for the most cautious, always to avoid collision! Many are the enactments made at different times in the different States of Flatland, in order to minimize this peril; and in the Southern and less temperate climates where the force of gravitation is greater, and human beings more liable to casual and involuntary motions, the laws concerning women are naturally more strict.

1. Every house shall have one entrance in the Eastern side, for the use of females only; by which all females shall enter "in a becoming and respectful manner" and not by the men's or Western door.
2. No female shall walk in any public place without continually keeping up her peace, under penalty of death.
3. Any female duly certified as invisible shall be allowed to wear a Dance, figs, chronic cold accompanied by violent sneezing,

1b/3 - 14% wife of a common Equilateral, who can achieve nothing beyond a mere monotonous swing, like the ticking of a pendulum; and the regular tick of the Equilateral is no less accorded and copied by the wife of the progressive and aspiring Isosceles, in the female of whose family "back motion" of any kind has become as yet a necessity of life. Hence, in every family of position and consideration, "back motion" is as prevalent as time itself; and the husbands and sons in these households enjoy immunity at least from invisible attacks. Not that it must be for a moment supposed that our women are destitute of affection. But unfortunately the passion of the woman predominates, in the frail sex, over every other consideration. In the female of our family, "back motion" is an unfortunate conformation. For as they have no pretensions to an angle, being inferior in this respect to the very lowest of the Isosceles, they are consequently wholly devoid of brainpower, and have neither reflection, judgment nor forethought, and hardly any memory. Hence, in their fits of fury, they remember no claims and recognize no distinctions. I have actually known a case where a woman has exterminated her whole household, and afterwards, when her rage was over, and the fragments swept away, has asked what has become of her husband and her children. Obviously then a woman is not to be irritated as long as she is in a position where she can turn round. When you have them in their apartments—which are constructed with a view to denying them that power—you can say and do what you like; for they are then wholly impotent for mischief, and will not remember a few minutes hence the name of the person who has done them wrong. Hence, when they are with death, nor the promises which you may have found it necessary to make in order to pacify their fury. On the whole we get on pretty smoothly in our domestic relations, except in the lower strata of the military classes. There the want of tact and discretion on the part of the husbands produces at times indescribable disasters. Relying too much on the offensive weapons of their acute angles instead of the defensive organs of good sense and seasonal simulation, these reckless

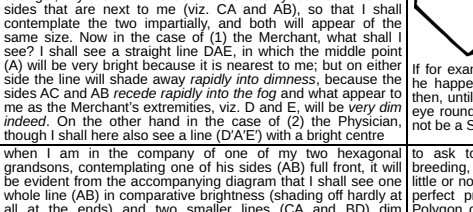
woman's mouth; and as the wife has absolutely nothing to say, and absolutely no constraint of wit, sense, or conscience to prevent her from saying it, not a few cynics have been found to aver that they prefer the danger of the death-dealing, and audible sting to the safe and insidious, and harmless, and often end. To my readers in Spaceland the condition of our women may seem truly deplorable, and so indeed it is. A male of the lowest type of the Isosceles may look forward to some improvement of his angle, and to the ultimate elevation of the whole of his degraded caste; but no woman can entertain such hopes for her sex. "Once a woman, always a woman" is a decree of Nature, and the very laws of Evolution seem to have decreed that the female of our species shall remain as she is. You see, dear reader, that the female of our species is a creature of a prearrangement which has ordained that, as they have no hopes, so they shall have no memory to recall, and no forethought to anticipate, the miseries and humiliations which are at once a necessity of their existence and the basis of the constitution of Flatland.

3. When I was in Spaceland I understood that some of your priestly circles have in the same way a separate entrance for villagers, farmers and teachers. Board Schools (Spectator, Sept. 1884, p. 1255) that they may "approach in a becoming and respectful manner." [Sheet 1 - Front - Cell 15]

V Of our Methods of Recognizing One Another

You, who are blessed with shade as well as light, you, who are gifted with two eyes, endowed with a knowledge of perspective, and charmed with the enjoyment of various colours, you, who can actually see an angle, and who are not hampered by the complete circumference of a circle in the happy region of the Three Dimensions—how shall I make clear to you the extreme difficulty which we in Flatland experience in recognizing one another's configuration? Recall what I told you above. All beings in Flatland, animate or inanimate, no matter what their form, present to our view the same, or nearly the same, appearance,

1b/3 - 23% transparent. But wherever there is a rich supply of fog, objects that are at a distance, say of three feet, are appreciably dimmer than those at a distance of two feet eleven inches; and the result is that by careful and constant experimental observation of comparative dimness and clearness, we are enabled to infer with great exactness the configuration of the object observed. An instance will do more than a volume of generalities to make my meaning clear. Suppose I see two angles, one of which I call A, and the other B. They are, we will suppose, a Merchant and a Physician, or in other words, an Equilateral Triangle and a Pentagon: how am I to distinguish them?

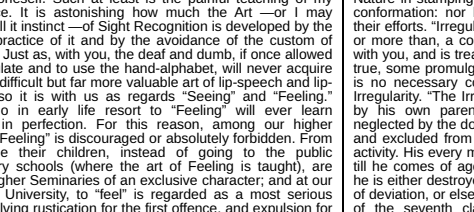


It will be obvious, to every child in Spaceland who has touched the threshold of Geometrical Studies, that, if I can bring my eye so that its glance may bisect an angle (A) of the approaching stranger, my view will lie as it were evenly between his two sides that are next to me (viz. CA and AB), so that I shall contemplate the two impartially, and both will appear of the same size. Now in the case of (1) the Merchant, what shall I see? I shall see the same angle, and I shall see the same line. The angle (A) will be very bright because it is nearest to me; but on either side the line will shade away rapidly into dimness, because the sides AC and AB recede rapidly into the fog and what appear to me as the Merchant's extremities, viz. D and E, will be very dim indeed. On the other hand in the case of (2) the Physician, though I shall here also see a line (D/AE) with a bright centre when I am in the company of one of my more hexagonal grandsons, contemplating one of his sides (AB) full front, it will be evident from the accompanying diagram that I shall see one side of the States. It is only a few of the scions of my noblest and wealthiest houses, who are able to give the time and money necessary for the thorough prosecution of this noble and valuable Art. Even to me, a mathematician of no mean standing, and the grandfather of two most hopeful and perfectly regular Hexagons, to find myself in the midst of a crowd of rotating Polygons of the higher classes, is occasionally very perplexing. And of course to a common tradesman, or serf, such a sight would be as much as to see the sun. Hence, when I, your reader, were you suddenly transported into our country, in such a crowd you could see on all sides of you nothing but a Line, apparently straight, but of which the parts would vary irregularly and perpetually in brightness or dimness. Even if you had completed your third year in the Pentagonal and Hexagonal classes in the University, and were perfect in the theory of the subject, you would still find that there was need of many years experience, before you could move in a fashionable crowd without justifying against your betters, whom it is against etiquette

toatures too often neglect the prescribed construction of the women's apartments, or irritate their wives by ill-advised expressions out of doors, which they refuse immediately to retract. Moreover a blunt and stolid regard for literal truth indisposes them to make those concessions by which the more judicious Circle can in a moment pacify his consort. The result is massacre; not, however, without its advantages, as it eliminates the more brutal and troublesome of the Isosceles; and by many of our Circles the destructiveness of the thinner sex is regarded as one among many providential arrangements for suppressing redundant population, and nipping Revolution in the bud. Let it never in your well regulated and most approximately Circular families, I cannot say that the ideal of family life is so high as with you in Spaceland. There is peace in so far as the absence of slaughter may be called by that name, but there is necessarily little harmony of tastes or pursuits; and the cautious wisdom of the Circles has ensured safety at the cost of domestic comfort. In every Circular or Polygonal household it has been a habit from time immemorial—and now has become a kind of instinct among the women—to make those concessions to the whims of their husbands which constantly keep their eyes and mouths towards their husband and his male friends; and for a lady in a family of distinction to turn her back upon her husband would be regarded as a kind of portent, involving loss of status. But, as I shall soon show, this custom, though it has the advantage of safety, is not without its disadvantages. In the house of the working man or respectable tradesman where the wife is allowed to turn her back upon her husband, while pursuing her avocations—there are at least intervals of quiet, when the wife is neither seen nor heard, except for the humming sound of the continuous peace-cry; but in the homes of the upper classes there is too often no peace. There the voluble mouth and bright penetrating eye are ever directed towards the master of the household; and light itself is not more persistent than the stream of feminine discourse. The tact and skill which suffice to avert a woman's sting are unequal to the task of stopping a

viz. that of a straight Line. How then can one be distinguished from another, where all appear the same? The answer is threefold. The first means of recognition is the sense of hearing; which with us is far more highly developed than with you, and which enables us not only to distinguish by the tone of our personal friends, but even to discriminate between different classes, at least so far as concerns the three lowest orders, the Equilateral, the Square, and the Pentagon—for of the Isosceles I take no account. But as we ascend in the social scale, the process of discrimination and being discriminated by hearing increases in difficulty, partly because voices are assimilated, partly because the faculty of voice-discrimination has, upon the whole, been somewhat deteriorated by time. And wherever there is any danger of imposture we cannot trust to this method. Amongst our lowest orders, the vocal organs are developed to a degree more than correspondent with those of hearing, so that an Isosceles can easily feign the voice of a Polygon, and, with some training, that of a Circle himself. A second method is therefore more commonly resorted to. Feeling is, among our women and lower classes—about our middle and upper classes it is not so—the principal test of recognition, at all events between strangers, and when the question is, not as to the individual, but as to the class. What therefore "introduction" is among the higher classes in Spaceland, that the process of "feeling" is with us. "Permit me to ask you to feel and be felt by my friend Mr. So-and-so"—is still, among the more old-fashioned of our country gentlemen in districts remote from towns, the customary formula for a Flatland introduction. But in the towns, and among the circles of business, the words "be felt by" are omitted and the sentence is abbreviated to, "Let me ask you to feel Mr. So-and-so"; although it is assumed, of course, that the "feeling" is to be reciprocal. Among our still more modern and dashing young gentlemen—who are extremely averse to superfluous effort and supremely indifferent to the purity of their native language—the formula is still further curtailed by the use of "to feel" in a

(A), yet it will shade away less rapidly into dimness, because the sides (A'C, A'B) recede less rapidly into the fog; and what appear to me the Physician's extremities, viz. D' and E', will be not so dim as the Merchant's extremities. The reader will probably understand from these two instances how—after a very long training supplemented by constant experience—it is possible for the well-educated classes among us to discriminate with fair accuracy between the middle and lowest orders, by the sense of sight alone. But it is not so with the grasped this general conception, so far as to conceive the possibility of it and not to reject my account as altogether incredible—I shall have attained all I can reasonably expect. Were I to attempt further details I should only perplex. Yet for the sake of the young and inexperienced, who may perchance infer—from the two simple instances I have given above, of the manner in which I should recognize my father and my sons by recognition by feeling, that I am unable to do so, I will now point out that in actual life most of the problems of Sight Recognition are far more subtle and complex.



If, for example, when my father, the Triangle, approaches me, he happens to present his side to me instead of his angle, then, until I have asked him to rotate, or until I have edged my eye round him, I am for the moment doubtful whether he may not be a Straight Line, or, in other words, a woman. Again, to ask to "feel," and who, by their superior culture and breeding, know all about your movements, while you know very little or nothing about theirs. In a word, to comport oneself with perfect propriety in Polygonal society, one ought to be a Polygon oneself. Such at least is the painful teaching of my experience. It is astonishing how much the Art—or I may almost call it instinct—of Sight Recognition is developed by the habitual practice of it and by the avoidance of the custom of "Feeling." Just as, with you, the deaf and dumb, if once allowed to gesticulate and to use the hand-alphabet, will never acquire the more difficult but far more valuable art of lip-speech and lip-reading so with us as the blind, if once permitted to "feel," None who in early life resort to "Feeling" will ever learn "Seeing" in perfection. For this reason, among our higher classes, "Feeling" is discouraged or absolutely forbidden. From the cradle their children, instead of going to the public elementary schools (where the art of Feeling is taught), are sent to higher Seminaries of an exclusive character; and to our illustrious University, to "feel" regarded as a most serious fault, involving rustication for the first offence, and expulsion for the second. But among the lower classes the Art of Sight Recognition is regarded as an unattainable luxury. A common tradesman cannot afford to let his son spend a third of his life in abstract studies. The children of the poor are therefore allowed to "feel" from their earliest years, and they gain thereby a precocity and an early vivacity which contrast at first most favourably with the inert, undeveloped, and listless behaviour of the upper classes. But when the time comes for the sons of the latter have at last completed their University course, and are prepared to put their theory into practice, the change that comes over them may almost be described as a new birth, and in every art, science, and social pursuit they rapidly overtake and distance their Triangular competitors. Only a few of the Polygonal class fail to pass the final test or leaving examination at the University. The condition of the unsuccessful minority is a pitiable one. Reduced to the rank of the lower classes, they are despised by the lower. They have neither the matured and

1b/3 - 19% technical sense, meaning, "to recommend-for-the-purposes-of-feeling-and-being-felt"; and at this moment the "slang" of polite or fast society in the upper classes sanctions such a barbarism as "Mr. Smith, permit me to feel Mr. Jones." The barren vertex of an acute-angled Isosceles is obvious to the tedious process that it would be with you, or that we find it necessary to feel right round all the sides of every individual before we determine the class to which he belongs. Long practice and training, begun in the schools and continued in the experience of daily life, enable us to discriminate at once by the sense of touch, between the angles of an equal-sided Triangle, Square, and Pentagon; and I need not say that the barren vertex of an acute-angled Isosceles is obvious to the duldest touch. It is therefore not necessary, as a rule, to do more than feel a single angle of an individual; and this, once ascertained, tells us the class of the person whom we are addressing, unless indeed he belongs to the higher sections of the nobility. There the difficulty is much greater. Even a Master of Arts in our University of Wentbridge has been known to confuse a ten-sided with a twelve-sided Polygon, and there have been cases of a nobleman mistaking a twenty-sided for a twenty-sided member of the aristocracy. Those of my readers who recall the extracts I gave above from the legislative code concerning women, will readily perceive that the process of introduction by contact requires some care and discretion. Otherwise the angles might inflict on the unwary Feeling irreparable injury. It is essential for the safety of the feeling that the hand should be held in a certain way, a fidgety shifting of the position, yes, even a violent sneeze, has been known before now to prove fatal to the incautious, and to nip in the bud many a promising friendship. Especially is this true among the lower classes of the Triangles. With them, the eye is situated so far from their vertex that they can scarcely take cognizance of what goes on at that extremity of their frame. They are, moreover, of a rough coarse nature, not sensitive to the delicate touch of the highly organized

at all) by half a degree in every generation; until the goal of 60° is reached, when the condition of serfdom is quitted, and the freeman enters the class of Regulars. Consequently, Nature herself supplies us with an ascending scale or alphabet of the sense of feeling, and a corresponding scale of education is placed in every elementary school throughout the land. Owing to occasional regressions, to still more frequent moral and intellectual stagnation, and to the extraordinary fecundity of the criminal and vagabond classes, there is always a vast superfluity of individuals of the half degree and single degree class, and a fair abundance of specimens up to 10°. These are absolutely destitute of civic rights, and a great number of them, especially those of the half degree and single degree class, are devoted by the States to the service of education. Fettered immovably so as to remove all possibility of danger, they are placed in the class rooms of our infant schools, and there they are utilized by the Board of Education for the purpose of imparting to the offspring of the middle classes that tact and intelligence of which these wretched creatures themselves are utterly devoid. In some States the specimens of the half degree and single degree class are put to work in the more temperate and better regulated regions, it is found in the long run more advantageous for the educational interests of the young, to dispense with food, and to renew the specimens every month—which is about the average duration of the foodless existence of the criminal class. In the cheaper schools, which is gained by the longer existence of the specimen is lost, partly in the expensure for food, and partly in the safety of the feeling. It is therefore to be expected that, after a few weeks of constant "feeling," Nor must we forget to add, in enumerating the advantages of the more expensive system, that it tends, though slightly yet perceptibly, to the diminution of the redundant Isosceles population—an object which every statesman in Flatland constantly keeps in view. On the whole therefore—although I am not ignorant that, in many popularly elected School Boards, there is a reaction in favour of "the

1b/3 - 27% systematically trained powers of the Polygonal Bachelors and Masters of Arts, nor yet the native precocity and mercantile versatility of the youthful tradesman. The professions, the public services, are closed against them; and though in most States they are not actually debarred from marriage, yet they have the greatest difficulty in forming suitable alliances, as experience shows that the offspring of such unfortunate and ill-endowed parents is generally itself unfortunate, and is doomed to exist several years; but in the more temperate and better regulated regions, it is found in the long run more advantageous for the educational interests of the young, to dispense with food, and to renew the specimens every month—which is about the average duration of the foodless existence of the criminal class. In the cheaper schools, which is gained by the longer existence of the specimen is lost, partly in the expensure for food, and partly in the safety of the feeling. It is therefore to be expected that, after a few weeks of constant "feeling," Nor must we forget to add, in enumerating the advantages of the more expensive system, that it tends, though slightly yet perceptibly, to the diminution of the redundant Isosceles population—an object which every statesman in Flatland constantly keeps in view. On the whole therefore—although I am not ignorant that, in many popularly elected School Boards, there is a reaction in favour of "the

VII Concerning Irregular Figures

Throughout the previous pages I have been assuming—that perhaps should have been laid down at the beginning as a distinct and fundamental proposition—that every human being in Flatland is a Regular Figure, that is to say of regular construction. But, as I have said, the sense of feeling is not a line, but a straight line; that an artisan or soldier must have two of his sides equal; that tradesmen must have three sides equal; Lawyers (of which class I am a humble member), four sides equal, and generally, that in every Polygon, all the sides must be equal. The size of the sides would of course depend upon the age of the individual. A female at birth would be about an inch long, while a tall adult woman might extend to a foot. As to the seventh class, prevented from marriage, forced to drudge at an uninteresting occupation for a miserable stipend; obliged to live and board at the office, and to take even his vacation under close supervision; what wonder that human nature, even in the best and purest, is embittered and perverted by such surroundings! All this very plausible reasoning does not convince me; as it has not convinced the wisest of our statesmen, that our ancestors erred in laying it on the shoulders of the lower classes to do the dirty work, incompatible with the safety of the State. Doubtless, the life of an Irregular is hard; but the interests of the greater number require that it shall be hard. If a man with a triangular front and a polygonal back were allowed to exist and to propagate a still more irregular posterity, what would become of the arts of life? Are the houses and doors and churches in Flatland to be altered in order to accommodate such monsters? Are our public edifices to be enlarged to make room for them? Would I deny that they allow him to enter a theatre or to take his

Figure in the company, all would be chaos and confusion, and the slightest panic would cause serious injuries, or—if it happened to be any woman, or soldiers present—perhaps a general massacre. Hence, the sense of feeling is not a line, but a straight line; that an artisan or soldier must have two of his sides equal; that tradesmen must have three sides equal; Lawyers (of which class I am a humble member), four sides equal, and generally, that in every Polygon, all the sides must be equal. The size of the sides would of course depend upon the age of the individual. A female at birth would be about an inch long, while a tall adult woman might extend to a foot. As to the seventh class, prevented from marriage, forced to drudge at an uninteresting occupation for a miserable stipend; obliged to live and board at the office, and to take even his vacation under close supervision; what wonder that human nature, even in the best and purest, is embittered and perverted by such surroundings! All this very plausible reasoning does not convince me; as it has not convinced the wisest of our statesmen, that our ancestors erred in laying it on the shoulders of the lower classes to do the dirty work, incompatible with the safety of the State. Doubtless, the life of an Irregular is hard; but the interests of the greater number require that it shall be hard. If a man with a triangular front and a polygonal back were allowed to exist and to propagate a still more irregular posterity, what would become of the arts of life? Are the houses and doors and churches in Flatland to be altered in order to accommodate such monsters? Are our public edifices to be enlarged to make room for them? Would I deny that they allow him to enter a theatre or to take his

Polygon. What wonder then if an involuntary toss of the head has ere now deprived the State of a valuable life! I have heard that my excellent grandfather—one of the least irregular of his unhappy Isosceles class, who indeed obtained, shortly before his death, the rank of a nobleman, and thereby, in consequence of Social Board for passing him into the class of the Equal-sided—often deplored, with a tear in his venerable eye, a miscarriage of this kind, which had occurred to his great-great-grandfather, a respectable working man with an angle or brain of 59° 30'. According to his account, my unfortunate ancestor, being afflicted with rheumatism, and in the act of being felt by a Polygon, by one sudden start accidentally transfixed the great man through the diagonal; and thereby, partly in consequence of his long imprisonment and degradation, and partly because of the moral shock which pervaded the whole of my ancestor's relations, threw back our family a degree and a half in their ascent towards better things. The result was that in the next generation the family brain was registered at only 58°, and next fell the lapse of five generations was the lost ground recovered, till a full 60° attained, and the Ascent from the Isosceles finally achieved. I have heard, too, that my grandfather, in the process of Feeling. At this point I think I hear some of my better educated readers exclaim, "How could you in Flatland know anything about angles and degrees, or minutes? We can see an angle, because we, in the region of Space, can see two straight lines inclined to one another; but you, who can see nothing but one straight line at a time, or at all events only a number of bits of straight lines all in one straight line—how can you ever discriminate angles, and much less degrees, or minutes, or different sizes?" I answer that though we cannot see angles, we can infer them, and this with great precision. Our sense of touch, stimulated by necessity, and developed by long training, enables us to distinguish angles far more accurately than your sense of sight, when unaided by a rule or measure of angles. Nor must I omit to explain that we have great natural helps. It is with us a law of Nature that the brain of the Isosceles class shall begin at half a degree, or thirty minutes, and shall increase (if it increases

cheap system" as it is called—I am myself disposed to think that this is one of the many causes in which expense is the true economy. But I must not allow questions of School Board politics to divert me from my subject. Enough has been said, I trust, to impress the spirit of feeling on the reader's mind, and to induce a process as might have been supposed; and it is obviously more trustworthy than Recognition by hearing. Still there remains, as has been pointed out above, the objection that this method is not without danger. For this reason many in the middle and lower classes, and all without exception in the Polygonal and Circular orders, prefer a third method, the description of which shall be reserved for the next section.

VI Of Recognition by Sight

I am about to appear very inconsistent. In previous sections I have said that all Figures in Flatland present the appearance of a straight line; and it was added or implied, that it is consequently impossible to distinguish by the visual organ between individuals of different classes: yet now I am about to explain to my Spaceland critics how we are able to recognize one another by the sense of sight. If however the reader will be so good as to turn to the section on the Recognition by Feeling is stated to be universal, he will find this qualification—"among the lower classes." It is only among the higher classes and in our temperate climates that Sight Recognition is practised. That this power exists in any regions and for any classes is the result of fog; which prevails during the greater part of the year in all parts save the torrid zones. That which is with you in Spaceland an unmixed evil, blotting out the landscape, and depressing the spirit, and rendering the sense of sight, is by us recognized as a blessing scarcely inferior to air itself, and as the nurse of arts and parent of sciences. But let me explain my meaning, without further eulogies on this beneficent element. If fog were nonexistent, all lines would appear equally and indistinguishably clear; and this is actually the case in those unhappy countries in which the atmosphere is perfectly dry and

speaking of the equality of sides, and it does not need much reflection to see that the whole of the social life in Flatland rests upon the fundamental fact that Nature wills all Figures to have their sides equal. If our sides were unequal our angles might be unequal. Instead of its being sufficient to feel, or estimate by sight, a single angle in order to determine the form of an individual, it would be necessary to ascertain each angle by the experiment of Feeling. But life would be too short for such a weary and unprofitable life. The sense of feeling is not a line, but a straight line; that an artisan or soldier must have two of his sides equal; that tradesmen must have three sides equal; Lawyers (of which class I am a humble member), four sides equal, and generally, that in every Polygon, all the sides must be equal. The size of the sides would of course depend upon the age of the individual. A female at birth would be about an inch long, while a tall adult woman might extend to a foot. As to the seventh class, prevented from marriage, forced to drudge at an uninteresting occupation for a miserable stipend; obliged to live and board at the office, and to take even his vacation under close supervision; what wonder that human nature, even in the best and purest, is embittered and perverted by such surroundings! All this very plausible reasoning does not convince me; as it has not convinced the wisest of our statesmen, that our ancestors erred in laying it on the shoulders of the lower classes to do the dirty work, incompatible with the safety of the State. Doubtless, the life of an Irregular is hard; but the interests of the greater number require that it shall be hard. If a man with a triangular front and a polygonal back were allowed to exist and to propagate a still more irregular posterity, what would become of the arts of life? Are the houses and doors and churches in Flatland to be altered in order to accommodate such monsters? Are our public edifices to be enlarged to make room for them? Would I deny that they allow him to enter a theatre or to take his

VIII Of the Ancient Practice of Painting

If my readers have followed me with any attention up to this point, they will not be surprised to hear that life is somewhat dull in Flatland. I do not, of course, mean that there are not battles, conspiracies, tumults, factions, and all those other phenomena which are so common to the life of a State. I mean that I deny that the strange mixture of the problems of feeling, and the

problems of mathematics, continually inducing conjecture and giving the opportunity of immediate verification, imparts to our existence a zest which you in Spaceland can hardly comprehend. I speak now from the aesthetic and artistic point of view. I have no doubt that the artist in Spaceland is, and artistically, very dull indeed. How can it be otherwise, when all one's prospect, all one's landscapes, historical pieces, portraits, flowers, still life, are nothing but a single line, with no varieties except degrees of brightness and obscurity? It was not always thus. Colour, if tradition speaks the truth, once for the space of half a dozen centuries or more, threw a transient splendour over the lives of our ancestors in the remotest ages. Some private individuals, I suppose, had at that fashion spread like wildfire, and having casually discovered the constituents of the simpler colours and a rudimentary method of painting, is said to have begun decorating first his house, then his slaves, then his father, his sons, and grandsons, lastly himself. The convenience as well as the beauty of the results commended themselves to all. Wherever Chromatistes—for by that name the most trustworthy authorities concur in calling him—he lived, he attracted respect. No one now needed to "feel" him; no one mistook his front for his back; all his movements were readily ascertained by his neighbours without the slightest strain on their powers of calculation; no one jostled him, or failed to make way for him; his voice was saved the labour of that exhausting utterance by which we colourless Squares and Pentagons are often forced to proclaim our individuality when we move abroad in the wilderness. Before a week was over, every Square and Triangle in the district had copied the example of Chromatistes, and only a few of the more conservative Pentagons still held out. A month or two found even the Dodecagons infected with the innovation. A year had not elapsed before the habit had spread to all but the

including their acute angle; the militia of the Equilateral Triangles tricoloured in red, white, and blue; the mauve, ultramarine, gamboge, and burnt umber of the Square artillerymen rapidly rotating near their vermilion guns; the red and glorious the sensualist development of the Pentagons and Hexagons careering across the field in their offices of surgeons, geometicians and aides-de-camp—all these may well have been sufficient to render credible the famous story how an illustrious Circle, overcome by the artistic beauty of the forces under his command, threw aside his marshal's baton and his royal crown, exclaiming that he henceforth exchanged them for the artist's pencil. How great and glorious the specimens were no longer used nor needed, and refusing to pay the customary tribute from the criminal classes to the service of education, waxed daily more numerous and more insolent on the strength of their immunity from the old burden which had formerly exercised the twofold wholesome effect of at once taming their brutal nature and thinning their excessive numbers. Year by year the soldiers and

IX Of the Universal Colour Bill

But meanwhile the intellectual arts were fast decaying. The Art of Sight Recognition, being no longer needed, was no longer practised; and the studies of geometry, statics, kinetics, and other kindred subjects, came soon to be considered superfluous, and fell into disrespect and neglect even at our University. The inferior Art of Feeling speedily experienced the same fate at the elementary schools. Then the Isosceles classes, asserting that the specimens were no longer used nor needed, and refusing to pay the customary tribute from the criminal classes to the service of education, waxed daily more numerous and more insolent on the strength of their immunity from the old burden which had formerly exercised the twofold wholesome effect of at once taming their brutal nature and thinning their excessive numbers. Year by year the soldiers and

2a/3 - 40% masterpiece of rhetoric, which occupied nearly a day in the delivery, and to which no summary can do justice. With a grave appearance of impartiality he declared that as they were now finally committing themselves to reform or innovation, it was desirable that they should take one last view of the perimeter of the whole subject, its defects as well as its advantages. Gradually introducing the mention of the dangers to the tradesmen, the professional classes and the gentlemen, he dwelt on the magnitude of the evil, and he exhorted them that, in spite of all these defects, he was willing to accept the bill if it was approved by the majority. But it was manifest that all, except the Isosceles, were moved by his words and were either neutral or averse to the bill. Turning now to the workmen he asserted that their interests must not be neglected, and that, if they intended to accept the Colour Bill, they ought at least to do so with full view of the consequences. Many of them, he said, were of the Regular Triangles; others anticipated for their children a distinction they could not hope for themselves. That honourable ambition would now have to be sacrificed. With the universal adoption of colour, all distinctions would cease; Regularity would be confused with Irregularity; development would give place to retrogression; the workman would in a few generations be degraded to the level of the military, or even the convict and slave. The great evil of the criminal classes was their number, that is to say the criminal classes, who were already more numerous than the workmen, and would soon outnumber all the other classes put together when the usual Compensative Laws of Nature were violated. A subdued murmur of assent ran through the ranks of the artisans, and Chromatistes, in alarm, attempted to step forward and address them. But he found himself compassed with guards and forced to remain silent while the Chief Circle for the time being was punished by a final appeal to the women, exclaiming that, if the Colour Bill passed, no marriage would henceforth be safe, no woman's honour secure; fraud, deception, hypocrisy would pervade every

classes was again restored. Needless to say that henceforth the use of colour was abolished, and its possession prohibited, even the utterance of any word denoting colour, except by the Criminals, was punished by the severest penalty. Only at our University in some of the very highest and most esoteric classes—which I myself have never been privileged to attend—it is understood that the sparing use of colour is still sanctioned for the purpose of illustrating some of the deeper problems of mathematics. But of this I can only speak from hearsay. Elsewhere in Flatland, colour is now nonexistent. The art of making is known to only one living individual. The Chief Circle for the time being was punished by a final appeal to the women, exclaiming that, if the Colour Bill passed, no marriage would henceforth be safe, no woman's honour secure; fraud, deception, hypocrisy would pervade every

XI Concerning Our Priests

It is high time that I should pass from these brief and discursive notes about things in Flatland to the central event of this book, my initiation into the mysteries of Space. That is my subject; all that has gone before is merely preface. For this reason I must omit many matters of which the explanation would not, I flatter myself, be without interest for my readers: as for example, our method of propelling and stopping ourselves, although desitute of motion, or the manner in which we are punished by the stone, or brick, although of course we have no hands, nor can we lay foundations as you can, nor avail ourselves of the lateral pressure of the earth; the manner in which the rain originates in the intervals between our various zones, so that the northern regions do not intercept the moisture from falling on the southern; the nature of our hills and mines, our trees and vegetables, our seasons and festivals; our alphabet and method of writing, adapted to our linear tablets; these and a hundred other details of our physical existence I must pass

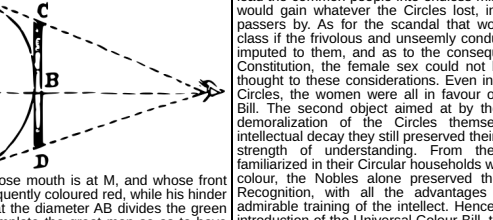
very highest of the nobility. Needless to say, the custom soon made its way from the district of Chromatistes to surrounding regions; and within two generations no one in all Flatland was colourless except the women and the priests. Here Nature herself had been the innovator, and the priests, by extending the innovation to these two classes. Many-sidedness was almost essential as a pretext for the innovators. "Distinction of sides is intended by Nature to imply distinction of colours"—such was the sophism which in those days flew from mouth to mouth, converting whole towns at a time to the new culture. But manifestly to our Priests and women this adage did not apply. The latter had only one side, and therefore—plurally and pedantically speaking—no sides. The former, if at least they would assert their claim to be really and truly Circles, and not mere high-class Polygons with an infinitely large number of infinitesimally small sides—were in the habit of boasting (what women confessed and deplored) that they also had no sides, being blessed with a perimeter of one line, or, in other words, a circumference. Hence it came to pass that these two classes could see no force in the so-called axiom about "Distinction of Sides in Nature." Distinction of colour and when it was introduced succumbed to the fascinations of corporal decoration, the Priests and the women alone still remained pure from the pollution of paint. Immoral, licentious, anarchical, unscientific—call them by what names you will—yet, from an aesthetic point of view, those ancient days of the Colour Revolt were the glorious childhood of art in Flatland—a childhood, alas, that never ripened into manhood, nor even reached the blossom of maturity. Even at a small party, the company was a pleasure to behold; the richly varied hues of the assembly in a church or theatre are said to have more than once proved too distracting for our greatest teachers and actors; but most ravishing of all is said to have been the unspeakable magnificence of a military review. The sight of a line of battle of twenty thousand Isosceles suddenly facing about, and exchanging the sombre blazon of their bases for the orange and purple of the two sides

artisans began more vehemently to assert—and with increasing truth—that there was no great difference between them and the very highest class of Polygons, now that they were raised to an equality with the latter, and enabled to graze on the same level of the social and when it was proposed whether statical or kinetical, by the simple process of Colour Recognition. Not content with the natural neglect into which Sight Recognition was falling, they began boldly to demand the legal prohibition of all "monopolizing and aristocratic arts" and the consequent abolition of all endowments for the studies of Sight Recognition, mathematics, and Feeling. Soon, they began to insist that inasmuch as Colour, which was a second nature, have wasted the need of a distinct colour, which was a second law should follow in the same path, and that henceforth all individuals and all classes should be recognized as absolutely equal and entitled to equal rights. Finding the higher orders wavering and undecided, the leaders of the Revolution advanced still further in their requirements, and at last demanded that all classes alike, the Priests and the women not excepted, should do homage to colour by submitting to be painted. When this was objected to, the Priests and the women, on their sides, they retorted that Nature and Expediency concurred in dictating that the front half of every human being (that is to say, the half containing his eye and mouth) should be distinguishable from his hinder half. They therefore brought before a general and extraordinary Assembly of all the States of Flatland a bill proposing that in every woman the half containing the eye and mouth should be coloured red, and the hinder half, which was the hind half, should be coloured green. In this way, red being applied to that semicircle in which the eye and mouth formed the middle point, while the other or hinder semicircle was to be coloured green. There was no little cunning in this proposal, which indeed emanated not from any Isosceles—for no being so degraded would have had angularity enough to appreciate, much less to devise, such a model of statecraft—but from an Irregular Circle who, instead

household; domestic bliss would share the fate of the Constitution and pass to speedy perdition. "Sooner than this," he cried, "come death." At these words, which were the preconcerted signal for action, the Isosceles Convicts fell on and transfixed the wretched Chromatistes; the regular classes, opening their ranks, made way for a band of women who, under direction of the Circles, moved, back foremost, invisibly and unannounced upon the unconscious soldiers; the artisans, maintaining the same position, also moved back foremost. Meaning bands of convicts occupied every entrance with an impenetrable phalanx. The battle, or rather carnage, was of short duration. Under the skillful generalship of the Circles almost every woman's charge was fatal and very many extracted their sting uninjured, ready for a second slaughter. But no second blow was needed; the rabble of the Isosceles died at the close of the business for themselves. Surprised, leaderless, and without the least idea of the utter ruin which was being wrought off by the convicts behind them, they at once—after their manner—lost all presence of mind, and raised the cry of "treachery." This sealed their fate. Every Isosceles now saw and felt a foe in every other. In half an hour not one of that vast multitude was living; and the fragments of seven score thousand of the criminal class in their one another's angles attest the triumph of Order. The Circles declined not to push their advantage to the uttermost. The slain were immediately decimated. The militia of the Equilaterals was at once called out; and every Triangle suspected of irregularity on reasonable grounds, was destroyed by court martial, without the formality of exact measurement by the Social Board. The homes of the military and artisan classes were inspected in a course of visitations extending through upwards of a year; and during that period every town, village, and hamlet was systematically inspected, and the least deviation from the prescribed order brought about by the neglect to pay the tribute of criminals to the schools and University, and by the violation of the other natural laws of the Constitution of Flatland. Thus the balance of

over, nor do I mention them now except to indicate to my readers that their omission proceeds not from forgetfulness on the part of the author, but from his regard for the time of the Constitution. The Circles declined not to push their advantage to the uttermost. The slain were immediately decimated. The militia of the Equilaterals was at once called out; and every Triangle suspected of irregularity on reasonable grounds, was destroyed by court martial, without the formality of exact measurement by the Social Board. The homes of the military and artisan classes were inspected in a course of visitations extending through upwards of a year; and during that period every town, village, and hamlet was systematically inspected, and the least deviation from the prescribed order brought about by the neglect to pay the tribute of criminals to the schools and University, and by the violation of the other natural laws of the Constitution of Flatland. Thus the balance of

2a/3 - 36% of being destroyed in his childhood, was reserved by a foolish indulgence to bring desolation on his country and destruction on myriads of his followers. On the one hand the proposition was calculated to bring the women in all classes on the side of the Chromatic Innovation. For by assigning to the women the same two colours as were assigned to the Priests, the Revolutionists thereby ensured that, in certain positions, every woman would appear like a Priest, and be treated with corresponding respect and deference—a prospect that could not fail to attract the female sex in a mass. But by some of my readers the possibility of the identical appearance of Priests and women, under the new legislation, may not be recognized; if so, a word or two will make it obvious. Imagine a woman duly decorated, according to the new code; with the front half (i.e. the half containing eye and mouth) red, and with the hinder half green. Look at her from one side. Obviously you will see a straight line, half red, half green.



Now imagine a Priest, whose mouth is at M, and whose front semicircle (AMB) is consequently coloured red, while his hinder semicircle is green; so that the diameter AB divides the green from the red. If you contemplate the great man so as to have your eye in the same straight line as his dividing diameter (AB), what you will see will be a straight line (CBD), of which one half

Irregular who I described above as the real author of this diabolical bill, determined at one blow to lower the status of the hierarchy by forcing them to submit to the pollution of colour, and at the same time to destroy their domestic opportunities of beauty, and to deprive them of the pleasure which new apparel and intellects by depriving them of their pure and colourless homes. Once subjected to the chromatic taint, every parental and every childish Circle would demoralize each other. Only in discerning between the father and the mother would the Circular infant find problems for the exercise of its understanding—problems too often likely to be corrupted by maternal impostures with the result of shaking the child's faith in all logical conclusions. Thus by degrees the intellectual lustre of the Priestly Order would wane, and the road would then lie open for a total destruction of all aristocratic legislature and for the subversion of our privileged classes.

X Of the Suppression of the Chromatic Sedition

The agitation for the Universal Colour Bill continued for three years; and up to the last moment of that period it seemed as though anxiety were destined to triumph. A whole army of death, who turned out to fight the private sinners, was utterly annihilated by a superior force of Isosceles Triangles—the Squares and Pentagons meanwhile remaining neutral. Worse than all, some of the ablest Circles fell a prey to conjurage fury. Infuriated by political animosity, the wives in many a noble household wearied their lords with prayers to give up their opposition to the Colour Bill; and some, finding their entreaties fruitless, fell on and slaughtered their innocent husbands, and turned their hands themselves in the act of carnage. It is recorded that during that triennial agitation no less than twenty-three Circles perished in domestic discord. Great indeed was the peril. It seemed as though the Priests had no choice between submission and extermination; when suddenly the course of events was completely changed by one of those picturesque incidents which statesmen often neglect to

2a/3 - 44% Circles in the social scale is not restricted, as it is among the lower Regular classes, by the law of Nature which limits the increase of sides to one in each generation. If it were so, the number of sides in a Circle would be a mere question of pedigree and arithmetic, and the four hundred and ninety-seventh descendant of an Equilateral Triangle would necessarily be a Polygon with five hundred sides. But this is not the case. Nature's law prescribes two antagonistic decrees which step in to help the process of the higher evolution. First, higher in the scale of development, so development shall proceed at an accelerated pace; second, that in the same proportion, the race shall become less fertile. Consequently in the home of a Polygon of four or five hundred sides it is rare to find a son; more than one is never seen. On the other hand the son of a five-hundred-sided Polygon has been known to possess five hundred and fifty, or even six hundred sides. Art is also in to help the process of the higher evolution. Physicians have discovered that the small and tender sides of an infant Polygon of the higher class can be fractured, and his whole frame reset, with such exactness that a Polygon of two or three hundred sides sometimes—by no means always, for the process is attended with serious risk—but sometimes overlaps two or three hundred generations, and as it were doubles at a stroke, the number of his progenitors and the number of his posterity. The more the doctrine is followed in this way, Scarcely one out of ten survives. Yet so strong is the parental ambition among those Polygons who are, as it were, on the fringe of the Circular class, that it is very rare to find a nobleman of that position in society, who has neglected to place his firstborn in the Circular Neo-Therapeutic Gymnasium before he has attained the age of a month. One year determines success or failure. At the end of that time the child is either put out of the question by the severity of the discipline in the Neo-Therapeutic Cemetery; but on rare occasions a glad procession bears back the little one to his exultant parents, no longer a Polygon, but a Circle, at least by courtesy;

and a single instance of so blessed a result induces multitudes of Polygonal parents to submit to similar domestic sacrifices, which have a dissimilar issue.

XII Of the Doctrine of Our Priests

As to the doctrine of the Circles it may briefly be summed up in a single maxim. "Attend to your Configuration." Whether political, ecclesiastical, or moral, all their teaching has for its object the improvement of individual and collective Configuration—with social and political progress the latter being the end, and the former which all other objects are subordinated. It is the merit of the Circles that they have effectually suppressed those ancient heresies which led men to waste energy and sympathy in the vain belief that conduct depends upon will, effort, training, encouragement, praise, or anything else but Configuration. It was Pantocylus—the illustrious Circle mentioned above, as the "queller of the Colour Revolt"—who first convinced mankind that the only way to secure the good of the State and of the family was to subordinate their own interests to those of posterity, thereby advancing the welfare of the whole State as well as that of their own immediate descendants. The weak point in the system of the Circles—if a humble Square may venture to speak of anything Circular as containing any element of weakness—appears to me to be found in their relations with women. As it is of the utmost importance for Society that Irregular birth should be suppressed, this is effected by carefully-kept pedigrees, which are preserved and supervised by the State; and without a certified pedigree no woman is allowed to marry. Now it might have been supposed that a Circle—proud of his ancestry and regardful for a posterity which might possibly issue hereafter in a Chief Circle—would be more careful than any other to choose a wife who had no blot on her escutcheon. But it is not so. The care in choosing a Regular wife appears to diminish as one rises in the social scale. The more the pedigree is ancient, the more the family had hopes of generating an Equilateral son, to take a wife who reckoned a single Irregularity among her ancestors; a Square or Pentagon, who is confident that his family is steadily on the rise, does not inquire above the five-hundredth generation; a Hexagon or Dodecagon is even more careless of the wife's pedigree; but a Circle has been known deliberately to take a wife who has had a regular great-grandfather, and all his children, and a slight superior pedigree, and who, by the charms of a low voice—which, with us, even more than you, is

(CB) will be red, and the other (BD) green. The whole line (CD) will be rather shorter perhaps than that of a full-sized woman, and will shade off more rapidly towards its extremities; but the identity of the colours would give you an immediate impression of identity. They anticipated your delight and confusion that would ensue. At home they might hear political and ecclesiastical secrets intended not for them but for their husbands and brothers, and might even issue commands in the name of a priestly Circle; out of doors the striking combination of red and green, without addition of any other colours, would be sure to lead the common people into endless mistakes, and the women would gain whatever the Priest lost, in the deference the masses would accord for the standard of the colours of the Circles, class if the frivolous and unseemly conduct of the women were imputed to them, and as to the consequent subversion of the Constitution, the female sex could not be expected to give a thought to these considerations. Even in the households of the Circles, the women were all in favour of the Universal Colour Bill. The second object aimed at by the bill was the gradual demoralization of the Circles themselves. In the general ecclesiastical and political teaching, the priests had the strength of understanding. From their earliest childhood, familiarized in their Circular households with the total absence of colour, the Nobles alone preserved the sacred art of Sight Recognition, with all the advantages that result from that admirable training of the intellect. Hence, up to the date of the introduction of the Universal Colour Bill, the Circles had not only held their own, but even increased their lead of the other classes by abstinance from the popular fashion. Now therefore the artful

neglect, often to anticipate, and sometimes perhaps to originate, because of the absurdly disproportionate power with which they appeal to the sympathies of the populace. It happened that an Isosceles of a low type, with a brain little if at all above four hundred sides, had been in the habit of visiting a certain tradesman whose shop he had plundered—painted himself, or caused himself to be painted (for the story varies) with the twelve colours of a Dodecagon. Going into the Market Place he accosted in a feigned voice a maiden, the orphan daughter of a noble Polygon, whose affection in former days he had sought in vain; and by a series of deceptions—aided, on the one side, by a string of lucky accidents too long to relate, and on the other, by the Isosceles' own sagacity—he succeeded in inducing her to take precautions on the part of the relations of the bride—he succeeded in consummating the marriage. The unhappy girl committed suicide on discovering the fraud to which she had been subjected. When the news of this catastrophe spread from State to State the minds of the women were violently agitated. Sympathy with the miserable victim and anticipations of similar deceptions for themselves, their sisters, and their daughters, soon drew round the Isosceles a large number of new converts. Not a few openly avowed themselves converted to antagonism; the rest needed only a slight stimulus to make a similar avowal. Seizing this favourable opportunity, the Circles hastily convened an extraordinary Assembly of the States; and besides the usual guard of convicts, they secured the attendance of a large number of reactionary women. Amidst an unprecedented concourse, the Chief Circle, those days by name Pantocylus, arose to read first himself and then to read by him a hundred and twenty thousand Isosceles. But he secured silence by declaring that henceforth the Circles would enter on a policy of concession; yielding to the wishes of the majority, they would accept the Colour Bill. The uproar being at once converted to applause, he invited Chromatistes, the leader of the Sedition, into the centre of the hall, to receive in the name of his followers the submission of the hierarchy. Then followed a speech, a

ardon me, my Lord," replied I; "but to my eye the appearance is as of an Irregular Figure whose inside is laid open to the view; in other words, methinks I see no Solid, but a Plane such as we infer in Flatland; only of an Irregularity which betokens some monstrous criminal, and the very sight of it is painful to my eyes." "Nay, my Sphere," replied the Lord of the Plane, "because you are not accustomed to light and shade and perspective; just as in Flatland a Hexagon would appear a Straight Line to one who has not the Art of Sight Recognition. But in reality it is a Solid, as you shall learn by the sense of Feeling." He then introduced me to the Cube, and I found that this marvelous being was indeed no Plane, but a Solid; and that the Sphere, in turn, was a Plane, and that the Plane was called solid angles; and I remembered the saying of the Sphere that just such a creature as this would be formed by a Square moving, in Space, parallel to himself; and I rejoiced to think that so insignificant a creature as I could in some sense be called the progenitor of so illustrious an offspring. But still I could not fully understand the meaning of what my teacher had told me concerning "light" and "shade" and "perspective"; and I did not know how to explain to the Sphere the things which he said. So the Sphere's explanation of these matters, succinct and clear though it was, it would be tedious to an inhabitant of Space, who knows these things already. Suffice it, that by his lucid statements, and by changing the position of objects and lights, and by allowing me to feel the several objects and even his own sacred person, he at last made all things clear to me, so that I could now readily distinguish between a Plane and a Sphere, a Solid and a Plane, and a Plane and a Solid, and I was able to follow my strange eventful history. Henceforth I have to relate the story of my miserable fall:—most miserable, yet surely most undeserved! For why should the thirst for knowledge be aroused, only to be disappointed and punished? My volition shrinks from the painful task of recalling my humiliation; yet, like a second Prometheus, I will endure this and worse, if by any means I may arouse in the interiors of Plane and Solid, if any

